**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR EXAMINATION APRIL 2026**

**SECOND SEMESTER MG2DSCFNN100– HOTEL HYGIENE AND FOOD SAFETY**

## PART A

1. Hazard Analysis and Critical Control Points.
2. Place a drop of milk on a slanted surface: pure milk flows slowly leaving a white trail. Adulterated milk flows quickly without a trace
3. Any substance consumed to provide nutritional support, energy, and growth to the body.
4. Cleaning: Removal of visible dirt and food particles, Sanitization: Reducing microorganisms to safe levels using chemicals or heat
5. Food hygiene refers to practices ensuring food safety from production to consumption. Importance: Prevents contamination, foodborne diseases, and ensures safe consumption.
6. Any biological, chemical, or physical agent in food that can cause harm to health.
7. Prevention of foodborne illness: Clean handling, Safe storage, Avoid cross-contamination, Personal hygiene, Proper cooking
8. Anti-microbial constituents effect
   They inhibit or kill microorganisms, slowing spoilage and preventing disease.
9. Yeast vs Mold (structural difference)
   - Yeast: Unicellular
   - Mold: Multicellular with filamentous structure (hyphae)
10. Water activity (aw)
    Measure of free water available for microbial growth (range: 0–1).
11. Relative humidity effect
    High humidity promotes microbial growth; low humidity inhibits it.
12. Two milk adulterants - Water, starch
13. Symptoms of salmonellosis
    Diarrhea, fever, abdominal cramps, vomiting
14. Harmful effect of intoxication
    Illness caused by toxins in food, leading to nausea, vomiting, and sometimes severe complications.
15. Biological adulteration
    Occurs when harmful microorganisms or foreign biological substances contaminate food.

## PART B

16. Factors determining food quality (any five)
    1. Appearance (color, texture) 2. Taste and flavor 3. Nutritional value 4. Freshness 5. Safety (free from contamination) 6. Shelf life
17. Health hazards of chemical contaminants

1. Pesticides: poisoning, cancer, 2. Heavy metals: neurological damage, 3. Food additives (excess): allergies, 4. Cleaning chemicals: toxicity, 5. Industrial pollutants: long-term health effects.

18. Causes of food infection

    1.Consumption of contaminated food, 2. Poor hygiene, 3. Improper cooking, 4. Cross-contamination, 5. Unsafe storage condition

19. Common foodborne microorganisms

- Bacteria: Salmonella, E. coli, Listeria
- Viruses: Norovirus, Hepatitis A
- Fungi: Molds producing mycotoxins
- Parasites: Giardia, Tapeworm

20. Risk factors influencing foodborne illness

    1.Temperature abuse, 2. Poor sanitation, 3. Infected food handlers, 4. Contaminated water, 5. Improper storage

21. Symptoms of foodborne illness: Nausea, vomiting, diarrhea, Fever, abdominal pain, weakness, dizziness, head ache, chills

Severity varies:

- Bacteria → severe diarrhea
- Viruses → vomiting
- Toxins → rapid symptom

## PART C

22. Types of food hazards & risks

- Chemical Hazards - Harmful substances (liquids, gases, fumes, dusts) that can cause damage when ingested.

  Examples: Pesticides, cleaning agents, acids, heavy metals
  Health Risks: Poisoning, Respiratory problems (asthma, lung damage), Skin irritation and burns, Long-term effects like cancer

- Physical Hazards - Environmental factors that can harm the body

  Examples: hair, nail, shells, glass chips, packaging materials
  Health Risks: choking, bleeding

- Biological Hazards - Living organisms or their products that can cause disease.

Examples: Bacteria, viruses, fungi, parasites
Health Risks:Infections and diseases, Food poisoning, Allergic reactions, Spread of epidemics

23. Shapes & structure of microorganisms

1. Bacteria(shapes)
   - Cocci (round) – e.g., Staphylococcus
   - Bacilli (rod-shaped) – e.g., E. coli
   - Spirilla (spiral) – e.g., Spirillum

Structures: Cell wall, Cytoplasm, Nucleus (in eukaryotes), Flagella (movement), Spores (resistance)

2. Virus: outer coat-capsid, nucleic acid,
3. Mold: multicellular, thread like structure, non motile,
4. Yeast: single celled, oval or spherical
5. Protozoa: unicellular

24. Preventive measures for foodborne pathogens

- Maintain personal hygiene
- Cook food thoroughly
- Store at correct temperatures
- Avoid cross-contamination
- Use safe water
- Follow HACCP principles
- Regular cleaning and sanitization